American Computer Science League

2022-2023 ● Contest 3: Short Problems ● Classroom Division

1. Boolean Algebra

Simplify this Boolean expression:

$$(A + B) + AB + B$$

A. B
B. A
C. AB
D. $A + B$
E. 1

2. Boolean Algebra

How many ordered pairs make the following Boolean expression TRUE?

$$(\overline{A}\, B)(A + \overline{B})$$

A. 0
B. 1
C. 2
D. 3
E. 4

3. Data Structures

Build the binary search tree for the following string:

K E N T U C K Y D E R B Y

List the nodes at depth 3 from left to right.

A. CKT
B. BDRU
C. CDTU
D. EY
E. Y

4. Data Structures

Given an initially empty stack and the following commands on the stack, what item will be popped next?

PUSH(“M”), PUSH(“E”), PUSH(“R”), X=POP(), X=POP(), PUSH(“C”), PUSH(“H”), X=POP(), X=POP(), PUSH(“A”), X=POP(), PUSH(“N”), PUSH(“T”), PUSH(“O”), PUSH(“F”), X=POP(), PUSH(“V”), X=POP(), PUSH(“E”), PUSH(“N”), X=POP(), X=POP(), PUSH(“I”), X=POP(), PUSH(“C”), PUSH(“E”), X=POP(), X=POP()

A. M
B. N
C. T
D. O
E. C

5. FSAs and Regular Expressions

Given the following regular expression, how many in the list do NOT satisfy it?

[a-m]*[^emtsy]*t.*y

entry	dotty	ditsy	crusty
gently	glinty	topiary	occipitally
documentary	mightily	intestacy	slapstick

A. 2
B. 3
C. 4
D. 5
E. 6

6. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A}(A + \overline{B}) + \overline{A}\,B$$

A. A
B. $\overline{A}$
C. $\overline{A}\,B$
D. $\overline{B}$
E. 1

7. Boolean Algebra

How many ordered triples make this expression TRUE?

$$\overline{(A\,\overline{B} + C)}\;\overline{(B\,\overline{C} + A)}$$

A. 1
B. 2
C. 4
D. 5
E. 7

8. Data Structures

Build a binary search tree for the following string:

P R E A K N E S S S T A K E S

List all of the nodes that have only a left child.

A. A, E, S
B. E, S, S
C. E, R, S
D. R, S, S
E. S, S, S

9. Data Structures

Given an initially empty queue and the following commands on the queue, what item will be popped next?

PUSH(“R”), PUSH(“O”), PUSH(“M”), X=POP(), X=POP(), PUSH(“E”), PUSH(“O”), X=POP(), X=POP(), PUSH(“A”), X=POP(), PUSH(“N”), PUSH(“D”), PUSH(“J”), PUSH(“U”), X=POP(), PUSH(“L”), X=POP(), PUSH(“I”), PUSH(“E”), X=POP(), X=POP(), PUSH(“T”), X=POP()

A. U
B. I
C. E
D. L
E. T

10. FSAs and Regular Expressions

Given the following regular expression:

[b-ps-z]*[aei][j-p]*

How many of the following dance names satisfy this expression?

waltz	tango	chacha	polka
samba	foxtrot	charleston	paso
salsa	rhumba	mambo	quickstep

A. 0
B. 1
C. 2
D. 3
E. 4